

SEDAR 33: Gulf of Mexico Gag and Greater Amberjack

Gag

Data Workshop

Life History

Stock Definition

Increased genetic sampling should provide more precise estimations of exchange rates within the Gulf basin and the Atlantic. As well, The LHW recommends continued application of otolith chemistry methods to evaluate the population structure and connectivity of gag.

Oceanographic modeling efforts are advancing (3-d models). Larval transport and modeling efforts need to be supported and associated with development of an Integrated Coastal Ocean Observing System (ICOOS). There is evidence for different transport and retention processes operating along the northeastern Gulf and west Florida shelf. Attention should be given to different oceanic forcing mechanisms particularly focusing on wind-driven upwelling and Loop Current intrusion differences north and south of about latitude 28°. Further exploration of potential larval contribution (interannual variation) from Campeche to US waters is needed.

For the purpose of learning more about exchange between basins, and as indicated in SEDAR 10, tagging studies should be coordinated between researchers in the Gulf (including Mexico) and south Atlantic, particularly with respect to adult size and depth. Additional acoustic tagging of mature gag may contribute to identification of additional spawning aggregation sites warranting protection. In particular more investigation of potential spawning habitats south of 28° latitude along the WFS is needed.

Age and Growth

Gag age samples are under- represented from the recreational sector. This remains a trend over time and more attention to recreational sampling is warranted.

Reader comparison statistics can now be incorporated as uncertainty in aging within the Stock Synthesis model. Estimates of standard deviation at age will be calculated and forwarded for review at the assessment workshop.

Further review of the aging macro (the assignment of final annual age) is needed to deal with the possibility of early annulus formation (e.g. before January 1st). Thus the age macro may need to include the means of age demotion for some individual gag.

Natural Mortality

1.) As in SEDAR 10, recommended ranges of M: (0.10 - 0.20).

2.) Continue to investigate age-varying M models and their appropriateness.

3.) LHW recommends further research into mortality rates of pre-spawning gag as they migrate from seagrass meadows to the offshore environment.

Reproduction

Maturity: Continue to gather histological samples to monitor change in maturity that may occur over time. Further examination needs to be made regarding how uncertainty in maturity can be treated within Stock Synthesis. A research recommendation is that formal decision tables be developed regarding the assignment of maturity based upon the raw histological readings for tropical/subtropical species. Changes to a decision table could be made in a standardized way to gauge the effect of uncertainty in models and for different species. The LHW recommends that this subject be presented at workshops or scientific meetings to raise awareness and develop consensus and best approaches.

Sex ratio, spawning fraction and fecundity: Promote collection of grouper reproductive samples via observer programs. Scientific observers working onboard commercial vessels will be able to sample gag in the round (prior to routine gutting) throughout the year. With improved field sampling, estimation of sex ratio needs to be made with better design or accounting of factors such as cohort effect (strong vs. weak year classes), location, gear and seasonal timing (pre-aggregation, spawning, and post-aggregation).

Sex transition and mechanism of sex change: Further review of the utility of secondary sexual traits (copperbelly pigmentation) is warranted: 1) incorporate the secondary sex field formally into TIP 2) provide training to port agents and 3) for longitudinal analysis develop means to account for changes in fishery selectivity and cohort effects.

Mating system: The LHW recommends further study of the particular type of mating system in gag (leks or harems). The distinction may depend on the particular type and amount of androgen produced (Shepherd et al 2013). An expectation is that leks would be more male biased as opposed to harems. As well, more information is needed on the timing and control of sex change in gag.

Form of reproductive potential: Because of questions about how the stock synthesis model can incorporate reproductive potential, the LHW recommends that three forms of reproductive potential be examined further at the Assessment Workshop given the data and reproductive traits reviewed at the data workshop 1) SSB-combined for male & female 2) SSB-female only and 3) SSB-eggs based upon annual fecundity.

Fertilization success: Research needs to be conducted on the consequences of sex ratio on fertility. The LHW recognizes that experiments on fertility would be difficult to conduct directly on such a large bodied species as gag (but see Rowe et al. 2004, 2008). Improved understanding of the gag mating system together with better designed field estimation of sex ratios may advance our understanding. Together with better field data, further genetic monitoring of Allee or inbreeding effects may yield much more insight on fertility and male reproductive success.

Conversion Factors

Continue to work on adoption of consistent standards across survey and data collection programs. Encourage programs collecting gag meristics to report fork length. Avoid use of Excel trend line function with some known statistical deficiencies in favor of more robust algorithms for solving equations.

Commercial Fishery Statistics

Landings

- Improved dockside sampling for catch composition
- Improved dealer reporting to species
- Historical literature research for historical landings

IFQ

- Investigate dealer influence on IFQ allocation usage through dealer IFQ surveys

Discard

- Most appropriate method for incorporation of IFQ data into discard estimations
- Most appropriate method for incorporation of IFQ data into discard size compositions
- Increased observer coverage.
- More representative observer coverage.

Recreational Fishery Statistics

- 1) Evaluate the technique used to apply sample weights to landings
- 2) Continue and expand fishery dependent at-sea observer surveys to collect discard information.
- 4) Track Texas commercial and recreational discards.
- 5) Estimate variances associated with the headboat program.
- 6) Evaluate existing and new methods to estimate historical landings.

Measures of Population Abundance

Expand the use of molecular genetics to identify the grouper larvae in SEAMAP samples that cannot be positively identified as gag grouper because diagnostic morphological characters are not yet developed.

The IWG made note that the delta-lognormal index may not be the most appropriate distribution with some of the data presented. However, the lack of adequate diagnostics for different distributions preclude their use. The recommendation is that additional work be done with these other distributions (i.e. Poisson, negative binomial) in order to fully vet the methodology.

A calibration study is needed between the FWRI/NMFS video survey and the UF diver survey (UVC). The standardized reef systems reported in SEDAR33-DW03 are well suited for rigorous calibration studies, which could also include other sampling methods.

An exploration of the effects of the IFQ on the fishery dependent indices, specially the commercial handline and longline is needed. During the workshop, fisherman indicated that since the implementation of the IFQ, there has been a drastic change in fisheries behavior. There is also the possibility that dealers can directly influence this behavior. The need is to find a way to incorporate these years into the overall timer series or a recommendation to split the time series when the IFQ began.

Further consideration of how to combine the data from the juvenile surveys, including perhaps revisiting the seagrass weighting approach as well as incorporating otolith microchemistry data on the relative contribution of estuaries to nearshore populations, may improve the YOY index.

Discard Mortality Rate

Future studies reporting discard mortality estimates should provide data tables that report the number of fish by discard condition (e.g. dead or alive), the number of fish by depth and by length bin, complete descriptions of gear (reel and hook type), and whether fish were properly vented. In addition, analyses of long-term mortality estimates from tag-recapture studies should account for survivorship and the effects of variable fishing effort over spatial and temporal scales.

Integrated Ecosystem Assessment

Harmful algal blooms

A top research priority is to assess the long-term effects of periodic HAB disturbances on the biomass, spatial distribution and age distribution of exploited reef fish species and their prey. These effects are likely to impact population viability and safe extraction rates, and could become very important to Gulf of Mexico fisheries management if climate change brings with it an increased frequency and severity of HAB events. Research should explore two avenues: retrospective analysis of reef fish biomass trends using historic HAB time series as drivers of mortality and recruitment, and future projections that challenge the current management practices under a schedule of increasing HAB disturbance (e.g., as informed by IPCC climate change scenarios). Priority should be given to spatially-explicit and/or stochastic simulation methods able to integrate, at minimum, the following features: fisheries effects, age structure, trophic dynamics, habitat, nutrient loading and HAB considerations.

Extending the Walter et al. (2013) red tide index forward will also be important for species affected by harmful algal blooms. This would involve bridging the SeaWIFS-MODIS gap between 2010 and 2011 to maintain continuity of satellite data and calibration of information.

Ecosystem modeling

No stock-recruitment relationship is specified in OSMOSE-WFS. Rather, recruitment levels in OSMOSE-WFS emerge from model simulations, and are dependent on the survival of eggs and larvae in relation to the predation process and to the amount of plankton available. Therefore, the development of OSMOSE-WFS #2 will be useful to obtain estimates of recruitment deviations for gag in the past, present and future that will be compared to estimates of recruitment deviations by the Connectivity Modeling System. Discussions would then be needed on how the outputs of both the Connectivity Modeling System and OSMOSE-WFS could be integrated into Stock Synthesis.

The IEA working group would benefit from another biophysical modeling system based on a Lagrangian framework, Ichthyop (Lett et al. 2008), in the future, to obtain estimates of recruitment deviations for gag and other species evaluated within the SEDAR process. The use of the Ichthyop model would be interesting to have several different perspectives on the issue of recruitment deviations.

Estimates of natural mortality

The following research topics would be useful for improving estimates of M:

- Within-model framework hypothesis testing of whether M applies to all ages equally or whether certain size/ages more vulnerable due to life history, location, and physiology
- Simulation modeling work to determine how best to model episodic mortality in stock assessments
- Development of forecasting methods to incorporate some probability of red tide occurring in the future, i.e. is there some autocorrelation to annual events, etc.
- Field or lab based studies of the effects of red tides on fish; for example, can fish sense *K. brevis* and do they move in response?
- Research on how mortality occurs; asphyxiation, bioaccumulation of toxins, etc.
- Does the pattern of recolonization of areas decimated by red tide occur through movement of adults or through settlement of juveniles.
- Collections of fish during red tide events, which would allow for the size/age selectivity of mortality to be determined, and might also allow for some minimum estimates of total mortality

Estimates of recruitment

Increased knowledge on the reproductive behavior or adult gag and biology of larval gag grouper would lead to better parameterization of larval transport models, and thus more accurate estimates of recruitment strength. Specifically, three major areas of uncertainty exist:

- The location of gag grouper spawning. While some sites have been well-documented (e.g., the Madison-Swanson Reserve) it is unknown whether spawning occurs in other locations along the West Florida Shelf. Collaborative projects with fishers would be particularly helpful in regards to identifying other potentially important spawning sites.
- The density and size of gag grouper eggs. Because transport patterns are highly sensitive to the vertical location of eggs in the first several days after release, more realistic parameterization of

particles in this initial stage would lead to more accurate estimates. In particular, knowledge on the densities of both fertilized and unfertilized eggs, and the timing of fertilization, would be useful.

The vertical distribution of gag larvae in the post-flexion stage. Because gag grouper have an extended pelagic larval duration (up to about 2 months), the fate of these larvae is largely determined by the depth layer in which they exist. Because grouper larvae are found in relatively low abundances in plankton tows, very little data exists on the vertical distributions of *Epinepheline* larvae in the pelagic environment, and in most cases the larvae are only identified to subfamily level. Increased sampling and identification of these larvae to species level will be important for understanding the vertical distributions of this species in the pelagic phase.

Assessment Process

1. Develop scientific survey to obtain reliable age/size composition data. This is needed, particularly as the composition data coming from the fisheries is substantially impacted by changing selectivity. This might be done with a handline survey of fixed sites. The idea would be not necessarily to get a random sample of the age composition but a reliable, relative estimate where selectivity can be assumed constant. An index would be nice, too.
2. Develop/Evaluate methods to maintain continuity of fishery-dependent indices in light of management regulations and ITQs.
3. Determine most appropriate methods to deal with changing selectivity in fisheries over time, particularly changing selectivity related to management actions or targeting of specific cohorts.
4. Evaluate most appropriate methods to deal with unreliable historic discard size composition data so that discard ratios can be reliably estimated.
5. Evaluate the size/age specific mortality effects of red tides on gag populations.

Review Workshop

Below, the RW Panel highlights research recommendations they feel should be emphasized, and provides new recommendations partly based on assessment methodology and results.

A. Research needs and new suggestions partly based on assessment methodology and results:

- (1) Research should be conducted for the most appropriate value of steepness to be used for Gag – either through across a range of species (e.g. Ram database) or use of a well-estimated value from a closely related stock or species.
- (2) If an appropriate fixed value for steepness is found, further research to explore the estimation of parameters currently fixed in the model, such as natural mortality.
- (3) Further work on improving selectivity parameters that are poorly estimated from the data available, or highly correlated with other model parameters.

(4) Need more work on whether it is best to use either female or sexes combines (more conservative). The combined was what the assessment panel recommended.

(5) More research on video survey methodology or increasing samples size, as there is concern as to why video estimates do not match other indices.

B. Recommendations to improve the SEDAR Process:

(1) Due to the inherent complexity of highly parameterized statistical catch at age models (i.e. stock synthesis) and the relative scarcity of expert users, the review panel recommends that each SEDAR assessment workshop panel include at least one nationally recognized expert in the model used (e.g. SS). This expert could participate in person or by electronic means and would greatly facilitate the review process.

(2) There is concern over a variety of issues that emerge as a result of the Assessment Workshop was exclusively performed via webinars. The Review Panel emphasizes the importance of face-to-face meetings for improving the model development during the assessment phase. The panel feels that many of the issues uncovered during the review process could have been avoided and may have enabled the assessment team to provide a more polished product for review and in the end resulting in the best model possible.

Greater Amberjack

Data Workshop

Life History

Natural Mortality

- Expand sampling in the commercial fishery to try and obtain larger/older individuals since most ages to date are from the recreational fishery.
- Use fishery-independent surveys to sample YOY greater amberjack over the entire first year of life.

Age

- Continue annual ageing workshops and reference collection exchanges among laboratories to standardize methods. As a group, decide how to deal with fish that form an opaque zone late in the year (i.e., to count last opaque zone or not).
- Due to the difficulty in distinguishing the first annulus from the core region, measurements should be taken on a subset of young-of-the-year to age one greater amberjack otoliths to use as a reference.
- Since there is large variation in length-at-age and Murie and Parkyn (2008) found a significant relationship between otolith weight and body weight, examine the relationship between otolith weight and age.

- Cross-reference trip tickets and log book data to Biological Sampling Database to complete spatial records (depth, grid, etc.) to allow for increased analysis of spatial demographics.
- Expand sampling of commercial and recreational spear landing and long-line landings, as these are under-represented in the dataset.
- Expand sampling in the Western Gulf of Mexico, in particular off Texas, as this region is under-represented in the dataset.
- A general recommendation of the LHW is to expand design-based fishery-independent sampling to elucidate regional (i.e., eastern and western GOM) and sub-regional differences in the demographics of greater amberjack.

Reproduction

- There is a lack of information on spawning frequency and fecundity with size and age for greater amberjack in the Gulf of Mexico. Given the observed differences in sexual maturity, peak spawning season, and potential growth differences between the South Atlantic and Gulf of Mexico stocks of greater amberjack, it should be a research priority to obtain information on spawning frequency and fecundity with size and age for Gulf of Mexico greater amberjack.
- Given that sex ratios are skewed to females for fish > 1 m fork length (Smith et al. 2013 SEDAR33-DW27), if release mortality is low (Murie and Parkyn 2013b SEDAR33-DW29), then a slot size limit could be explored as a means of rebuilding female SSB.

Movement and Migration

- More tagging information is necessary to understand seasonal movements of greater amberjack in the Gulf of Mexico (see Stock ID section). Satellite tags may provide better habitat and seasonal information compared to conventional dart tags that cannot provide serial location information on the fish throughout the year.

Commercial Fishery Statistics

Landings

- Improved dockside sampling for catch composition
- Improved dealer reporting to species

Discards

- Increased observer coverage.
- More representative observer coverage.
- Most appropriate method for incorporation of IFQ data into discard estimations

Recreational Fishery Statistics

- 1) Evaluate the technique used to apply sample weights to landings.
- 2) Develop methods to identify angler preference and targeted effort.
- 3) Continue and expand fishery dependent at sea observer surveys to collect discard information. This would help to validate self-reported headboat discard rates.

- 4) Track Texas commercial and recreational discards.
- 5) Evaluate existing and new methods to estimate historical landings

Measures of Population Abundance

- Expand the use of molecular genetics to identify the amberjack larvae in SEAMAP samples that cannot be positively identified as greater amberjack because diagnostic morphological characters are not yet developed.
- The IWG made note that the delta-lognormal index may not be the most appropriate distribution with some of the data presented. However, the lack of adequate diagnostics for different distributions preclude their use. The recommendation is that addition work be done with these other distribution (i.e. Poisson, negative binomial) in order to fully vet the methodology.
- A calibration study is needed between the FWRI/NMFS video survey.
- An exploration of the effects of the IFQ on the fishery dependent indices, specially the commercial handline and longline is needed. During the workshop, fisherman indicated that since the implementation of the IFQ, there has been a drastic change in fisheries behavior. There is also the possibility that dealers can directly influence this behavior. The need is to find a way to incorporate these years into the overall timer series or a recommendation to split the time series when the IFQ began.

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Assessment Process

1. Review fishery dependent length and age sampling intensity protocols for Greater Amberjack. This is needed to optimize sampling coverage across the entire geographical area of catch.
2. Review fishery independent video surveys sampling design to determine if there are practical changes which could be implemented that would increase reliability in the indices. In particular, the Panama City trap video survey should be enhanced as this survey provides information on small Greater Amberjack. Improvements in the index could potentially yield more reliable estimates of size composition of recruits.
3. Develop fishery independent sampling programs for size/age composition. This research is needed to improve more reliable and accurate estimation of selectivity unaffected by fishery dependent data collections, the latter which are affected by management regulations
4. Evaluate method used to develop historical recreational effort.

5. Develop program/procedures to allow increased sampling of discarded fish for all fleets and initiate a program to collect size composition of discards from the private angler fleets. A program similar to the North Carolina Division of Marine Resources (i.e., the “Board Survey”) used to obtain size composition of discarded recreational fish) could be evaluated to obtain self-reported size composition from private anglers and other recreational components also.

Review Workshop

Below, the RW Panel highlights research recommendations they feel should be emphasized, and provides new recommendations partly based on assessment methodology and results.

A. Panel recommendations for other research needs and new suggestions partly based on assessment methodology and results:

1. Need more assessment analyses to determine whether it is best to use either female or sexes-combined biomass estimates
2. Improving discard mortality estimates should be considered
3. Species identification has the potential to be problematic. More studies using genetic approaches may be beneficial

B. Panel recommendations to improve the SEDAR Process:

1. Due to the inherent complexity of highly parameterized statistical catch at age models (i.e. stock synthesis) and the relative scarcity of expert users, the review panel recommends that each SEDAR assessment workshop panel include at least one nationally recognized expert in the model used. This expert could participate in person or by electronic means and would greatly facilitate the review process.

There is concern over a variety of issues that emerge as a result of the Assessment Workshop largely or even exclusively performed via webinars. The Review Panel emphasizes the importance of face-to-face meetings for improving the model development during the assessment phase. The panel feels that many of the issues uncovered during the review process could have been avoided and may have enabled the assessment team to provide a more polished product for review and in the end resulting in the best model possible.